KULDEEP PRAJAPAT

Ph.D. Scholar | School of Physical Sciences, IIT Mandi
Theoretical and computational atomic, molecular, and optical physics

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Google Scholar | GitHub

RESEARCH PROFILE

I study the structural and ultrafast electronic dynamics of atomic clusters, fullerenes, and endohedral systems. My interests include photoionization, electron correlation, collective excitations, interatomic Coulombic decay (ICD), plasmonic resonant ICD, photoelectron angular distributions, and Wigner-Smith time delays.

RESEARCH INTERESTS

- Photoionization dynamics in plasmonic metal clusters and fullerenes.
- Angular anisotropy and confinement resonances in A@C60 systems.
- Attosecond photoemission time delays in metal clusters, fullerenes, and endohedral systems.

EDUCATION AND CURRENT POSITION

Ph.D. in Physics | Indian Institute of Technology Mandi

School of Physical Sciences, Himachal Pradesh, India | August 2022-present | Advisor: Prof. Hari Varma R | Research area: atomic and molecular physics.

M.Sc. in Physics | Singhania University

Jhunjhunu, Rajasthan, India | 2018-2020.

CURRENT RESEARCH DIRECTIONS

- Angular-distribution studies of atomic clusters.
- Resonant plasmonic ICD effects on photoelectron angular distributions in Mg@C60 and Ar@C60.
- Time-delay investigations in photoionization of metal clusters and endohedral systems.

RESEARCH METHODS

Time-dependent density functional theory (TDDFT); photoelectron angular distributions (PADs); Wigner-Smith photoionization time delays; classical-trajectory Monte Carlo (CTMC) simulations.

TEACHING EXPERIENCE

- Theory of Atomic Collisions and Spectroscopy | Instructor: Prof. P. C. Deshmukh, IIT Tirupati.
- Theory of Quantum Collision and Spectroscopy | Instructor: Prof. Hari Varma, IIT Mandi.
- Classical Mechanics | Instructor: Prof. Hari Varma, IIT Mandi.
- Experimental Physics Lab | Instructor: Assoc. Prof. Pradeep Kumar, IIT Mandi.

TECHNICAL SKILLS

Programming: Fortran, Julia, Python, C | **HPC:** Param Himalaya at IIT Mandi
Software: Microsoft Office, LibreOffice | **Platforms:** Linux, Windows, macOS

MANUSCRIPTS IN PREPARATION

Single-electron photoexcitation ICD resonances riding with plasmons in Na20@C240

Kuldeep Prajapat, Hari Varma R, and Himadri Chakraborty | Preparing for submission.

Angle- and energy-resolved photoelectron distribution of C60 versus C240 fullerene

Kuldeep Prajapat, Hari Varma R, and Himadri Chakraborty | Manuscript in preparation.

Photoelectron angular-distribution studies of Na20 and Na92 clusters

Kuldeep Prajapat, Hari Varma R, and Himadri Chakraborty | Manuscript in preparation.

POSTERS

Attosecond Chronoscopy of Plasmon-Assisted Ionization Dynamics in Mg@C60

WUFXS2026: The 2026 International Workshop on Ultra-Fast & X-Ray Science, East China Normal University, Shanghai, China | April 16-19, 2026.

Angle- and Energy-Resolved Photoelectron Distribution of C60 versus C240 Fullerene

IAMNOP-2025: International Conference on Atomic, Molecular, Nano, and Optical Physics with Applications, Jawaharlal Nehru University, New Delhi, India | December 17-19, 2025.

Fano analysis of single-electron ICD resonances in plasmonic and non-plasmonic energy regions of Na20@C240

NCAMP 2025: 24th National Conference on Atomic and Molecular Physics, IIT Dhanbad, Jharkhand, India | January 8-11, 2025.

CONFERENCE PRESENTATIONS

Photoelectron angular-distribution studies of Na20 and Na92 clusters

9th ISAMP Topical Conference, PRL Ahmedabad, Gujarat, India | February 15-17, 2024.

Wigner time delay in the spillover plasmon resonance region of Na clusters

DAMOP 2023: 54th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics, Washington, USA | June 5-9, 2023.

Resonant intercluster Coulombic decay in the photoionization of Na20 confined inside C240

ICPEAC 2023: XXXIII International Conference on Photonic, Electronic and Atomic Collisions, Ontario, Canada | July 25-August 1, 2023.

QUALIFICATION AND ACADEMIC SERVICE

- Qualified GATE 2022.
- Organizing member, 67th Congress of the Indian Society of Theoretical and Applied Mechanics (ISTAM), IIT Mandi | December 14-16, 2022.
- Organizing member, In-House Symposium, IIT Mandi | May 9-10, 2023.

LANGUAGES

English and Hindi.

VERIFIED PUBLICATION RECORD

For the verified publication record and citation details, see Google Scholar.